

| Question  | Answer                                                                                                                                                                                                                                                                                                                                                                                                     | Marks    |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3(c)(iii) | <ul style="list-style-type: none"> <li>A <u>parity bit</u> is added to each <b>byte</b>.</li> <li>... to make the number of 1s/0s even // that will be 1 if the number of 1s/0s is odd // that will be 0 if the number of 1s/0s is even.</li> <li>The number of 1s/0s in each byte is counted <b>after transmission</b>.</li> <li>If any bytes have an odd number of 1s/0s an error is detected</li> </ul> | <b>4</b> |

| Question                                | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks     |             |                           |                                                                                               |                                         |                                                                    |       |                                                    |                      |                                                                    |              |                                                                                            |                                    |                                                                                                           |   |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------|---------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------|--------------------------------------------------------------------|-------|----------------------------------------------------|----------------------|--------------------------------------------------------------------|--------------|--------------------------------------------------------------------------------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------|---|
| 4(a)                                    | <p><b>One</b> mark for each correct component or description</p> <table><tr><th>Component</th><th>Description</th></tr><tr><td><b>Control unit // CU</b></td><td>It sends signals to all the components in the CPU to manage the flow of data through the CPU.</td></tr><tr><td><b>Arithmetic and logic unit // ALU</b></td><td>It carries out all the arithmetic and logic operations in the CPU.</td></tr><tr><td>Cache</td><td><b>It stores frequently used data/instructions</b></td></tr><tr><td>Program counter (PC)</td><td><b>It stores the address of the next instruction to be fetched</b></td></tr><tr><td><b>Clock</b></td><td>It controls the number of fetch-decode-execute (FDE) cycles that are performed per second.</td></tr><tr><td><b>Memory data register // MDR</b></td><td>It stores data immediately before it is transmitted to RAM and immediately after it is received from RAM.</td></tr></table> | Component | Description | <b>Control unit // CU</b> | It sends signals to all the components in the CPU to manage the flow of data through the CPU. | <b>Arithmetic and logic unit // ALU</b> | It carries out all the arithmetic and logic operations in the CPU. | Cache | <b>It stores frequently used data/instructions</b> | Program counter (PC) | <b>It stores the address of the next instruction to be fetched</b> | <b>Clock</b> | It controls the number of fetch-decode-execute (FDE) cycles that are performed per second. | <b>Memory data register // MDR</b> | It stores data immediately before it is transmitted to RAM and immediately after it is received from RAM. | 6 |
| Component                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |           |             |                           |                                                                                               |                                         |                                                                    |       |                                                    |                      |                                                                    |              |                                                                                            |                                    |                                                                                                           |   |
| <b>Control unit // CU</b>               | It sends signals to all the components in the CPU to manage the flow of data through the CPU.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |             |                           |                                                                                               |                                         |                                                                    |       |                                                    |                      |                                                                    |              |                                                                                            |                                    |                                                                                                           |   |
| <b>Arithmetic and logic unit // ALU</b> | It carries out all the arithmetic and logic operations in the CPU.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |           |             |                           |                                                                                               |                                         |                                                                    |       |                                                    |                      |                                                                    |              |                                                                                            |                                    |                                                                                                           |   |
| Cache                                   | <b>It stores frequently used data/instructions</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |           |             |                           |                                                                                               |                                         |                                                                    |       |                                                    |                      |                                                                    |              |                                                                                            |                                    |                                                                                                           |   |
| Program counter (PC)                    | <b>It stores the address of the next instruction to be fetched</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |           |             |                           |                                                                                               |                                         |                                                                    |       |                                                    |                      |                                                                    |              |                                                                                            |                                    |                                                                                                           |   |
| <b>Clock</b>                            | It controls the number of fetch-decode-execute (FDE) cycles that are performed per second.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |           |             |                           |                                                                                               |                                         |                                                                    |       |                                                    |                      |                                                                    |              |                                                                                            |                                    |                                                                                                           |   |
| <b>Memory data register // MDR</b>      | It stores data immediately before it is transmitted to RAM and immediately after it is received from RAM.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |           |             |                           |                                                                                               |                                         |                                                                    |       |                                                    |                      |                                                                    |              |                                                                                            |                                    |                                                                                                           |   |